Why Climate Disasters Don't Guarantee Climate Action?

Linking Regional Extreme Weather Events and Opinion Dynamics to Macroeconomic Outcomes in DSK+OD ABM

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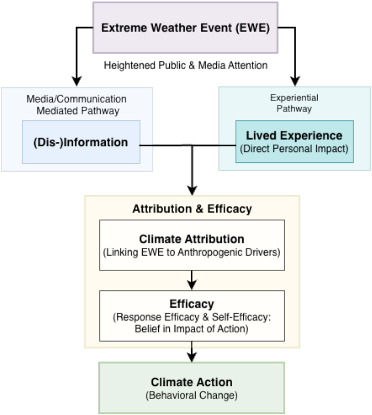
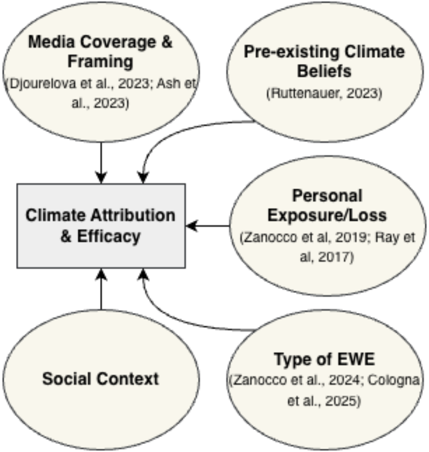
Introduction

- Extreme weather events (EWEs) are increasing in frequency, intensity and economic damages
  Intergovernmental Panel On Climate Change (2014); Coronese et al. (2019)
- Public and media attention are heightened during EWE
  Roxburgh et al. (2019); Friedman et al. (2019)
- EWEs can serve as catalysts for major societal and political shifts ("critical junction") Casali et al. (2024)
- Attribution of EWEs to climate change is key for converting experience into mitigation motivation Ogunbode et al. (2019)



What shapes climate attribution post EWE?



Regional heterogeneity in climate impacts & opinions

- Climate damages from EWEs are **uneven across regions**, creating long-run path dependencies in economic performance and climate policy support
  Ray et al. (2017); Giordono et al. (2020a)
- Regional EWE exposure differs in frequency as well as type
- National support for climate policy reflects interactions across regions, including:
  - **Opinion dynamics** among affected residents (e.g., climate attribution after shocks)
    Cologna et al. (2025)
  - **Economic & demographic weight** of impacted regions (core vs periphery)
    Bølstad (2015)
- Heterogeneous exposure & perceptions help explain **variation in local support for climate policies**
  Giordono et al. (2020b)

Research Question

How do regionally heterogeneous extreme weather events shape macroeconomic dynamics and climate policy outcomes through opinion dynamics?

1. How does heterogeneous regional exposure to extreme weather events affect regional economic trajectories and aggregate national macroeconomic outcomes?

   Phase 1

2. How do localized experiences of extreme weather events influence regional and national climate policy preferences, including both the strengthening and erosion of support for mitigation and adaptation policies?

   Phase 2

Model Extension

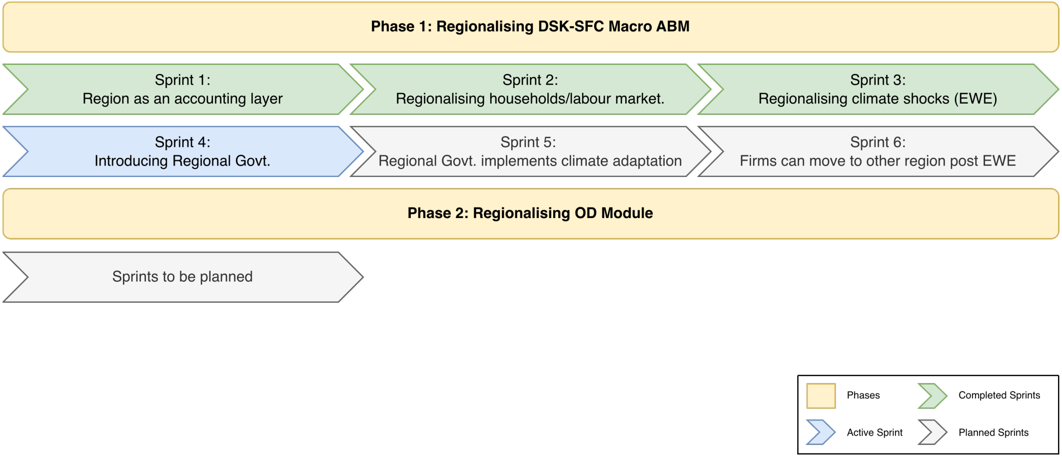
Phase 1: Regionalised Agents and Climate Shocks

- **Objective:** Explore how regional EWE exposures affect national and regional macro-economic outcomes.
- **Modelling:**
  - Regionalised agents: firms, labour (households), and energy plants
  - Capital Market, Goods Market and Energy Market remains at national level (unchanged).
  - Regional governments with budgets and policies.
- **Hypothesis:** Regional EWE damages create persistent economic path dependencies.
- **Policies:**
  - Baseline: no active regional governments.
  - Adaptation: regions offer relief subsidies or invest in damage reduction/rebuilding post EWE.
  - Policy mixes: test interactions of national mitigation and regional adaptation.

Phase 2: Opinion Dynamics and Regional Climate Shocks

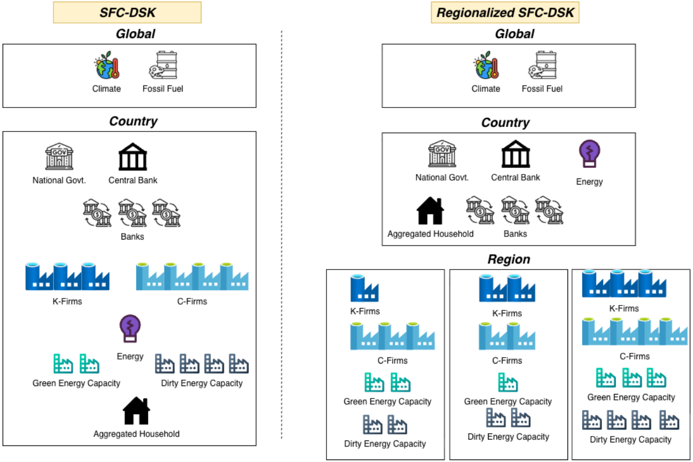
- **Objective:** Endogenize support for climate policies via opinion dynamics.
- **Modelling:**
  - Drivers: EWE exposure, attribution, beliefs, employment, (dis)information.
  - Outcomes: Differentiated support for mitigation (national) vs. adaptation (regional) policies.
- **Hypotheses:**
  - EWE exposure increases adaptation support; mitigation support is conditional see figure
  - Repeated EWE can either exacerbate or alleviate political polarization and climate fatalism among residents.
- **Policies:**
  - Baseline: Households form opinions on national carbon policy.
  - Adaptation scenario: Opinions include regional adaptation support.
  - Election scenario: Opinion shifts only affect policies during election cycles.

Model Extension Roadmap: Sprint Planning



Regional Firms

Sprint 1: Regionalising K & C Firms and Energy Capacity



Sprint 1: Regionalising K & C Firms and Energy Capacity

1. Region is conceptualized as an aggregator/accounting layer in the existing model. It is not an agent and does not influence model behaviour.
2. In this first sprint, K-Firms, C-Firms are tagged to a region.
3. Energy capacity, production, emissions, etc. are computed at regional level as derivatives based on regional capacity share.
4. ‘`AGGREGATE_REGIONS`’ function is introduced which reports regional aggregates and also act as a verification to ensure regional aggregates sum to global aggregates.
5. Other variables and behavioural aspects are left untouched.

Sprint 1: Regionalising K & C Firms

- If "`regions`" entry is present in the input file, firms are assigned a regional tag at initialization (sequential attribution).
- Upon exit and entry, firms inherit the regional tag associated with the slot they occupy.
- Assumption the proportion of K & C-firms does not evolve endogenously over the time.
- Assumption The market remains at national level. **No** regional dynamics here!

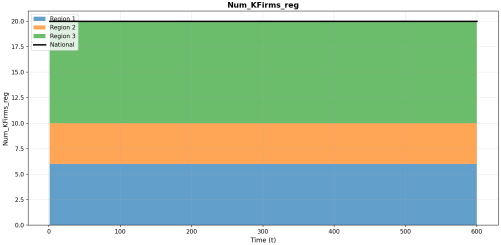


Figure: Number of K-Firms

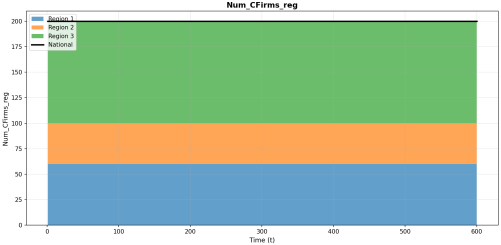


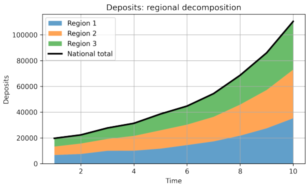
Figure: Number of C-Firms

Sprint 1: Regionalising Energy Capacity (Plants)

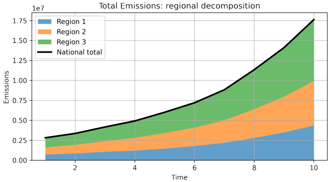
- Regional accounting layer computes following energy sector parameters (derived) at regional level using `dirty_capacity_share` and `green_capacity_share`.
- The energy market is also at national level analogous to a national electricity grid.
- The climate shock to energy capacity (plants) can be implemented using `dirty_capacity_share` and `green_capacity_share` when a region is hit by an EWE/Shock.

Sprint 1: Verifying Regional Aggregates (Sanity Check)

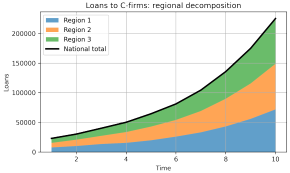
A 10-step simulation with three regions is executed to ensure that regional aggregates align with national-level aggregates.



Deposits



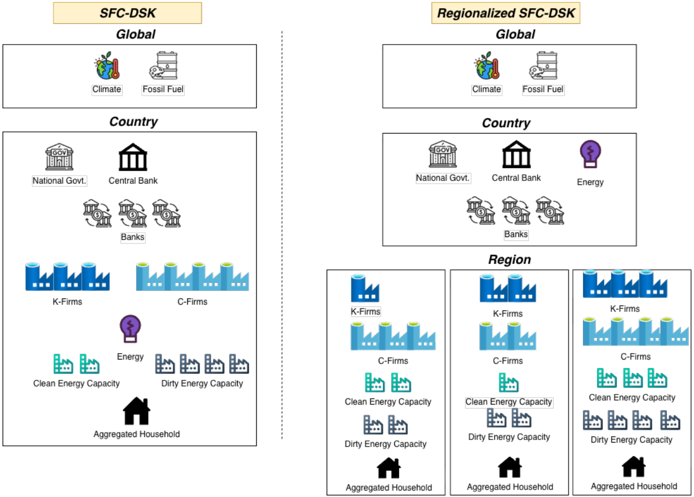
Emissions



loans

Regional Labour

Sprint 2: Regional Labour and Refining Regionalisation of Energy Capacity, K & C Firms



Sprint 2: Refining regional allocation of firms and energy capacity

- Introduced proportion-based regional allocation for:
  1. K-firms (static)
  2. C-firms (static)
  3. Dirty energy capacity & expansion
  4. Clean energy capacity & expansion
- Endogenous regional distribution of Labour ‘LS0‘
- Regional computation of Emissions from K-Firms, C-Firms & Energy sector

Excerpt from ‘regions_input.json‘

```
...
"regions":
{
  "NR": 3,
  "K_firm_shares": [0.3, 0.2, 0.5],
  "C_firm_shares": [0.3, 0.2, 0.5],

  "energy": {
    "dirty_capacity_shares": [0.1, 0.2, 0.7],
    "green_capacity_shares": [0.5, 0.3, 0.2]},

  "de_growth_probability": [0.1, 0.2, 0.7],
  "ge_growth_probability": [0.5, 0.3, 0.2]
}
...
```

Sprint 2: Probability based regional allocation for new energy capacity

- Distribution of clean and dirty energy capacity is specified in the `'regions_input.json'`
- New capacity expansion is determined based on regional growth probability specified in `'regions_input.json'`.

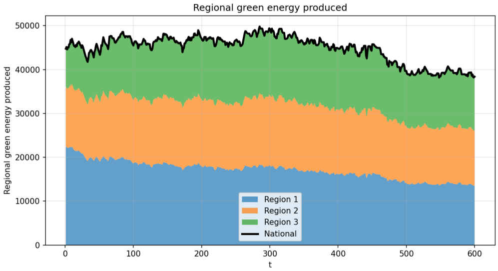


Figure: Green Energy Production

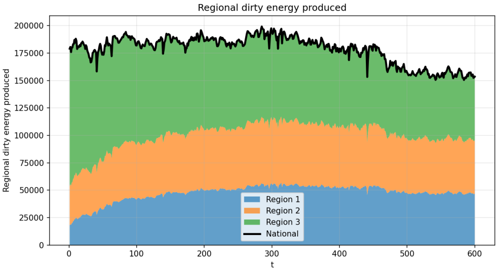


Figure: Dirty Energy Production

Sprint 2: Regionalising Households/Labour

1. Regional labour demand (C-Firms, K-Firms and Energy Sector) is computed during initialisation and Labour `'LS0'` is distributed across regions in a way that all labour demands are met at `t = 1`.
2. Regional labour shares are distributed endogenously based on labour demand by K and C firms in that region. This keeps the existing SFC-DSK mechanisms intact.
3. Assumption **Free inter-region mobility** If a labour loses employment, it will move to another region with labour demand.
4. Labour stays unemployed in a region, until a new labour demand emerges in the same or different region.

Sprint 2: Regionalising Households/Labour

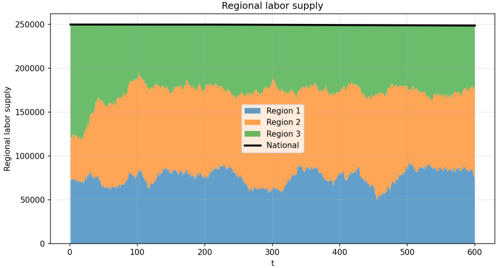


Figure: Labour Supply Pool LS0

Figure: Employment Rate

Sprint 2: Regionalising Firm Emissions

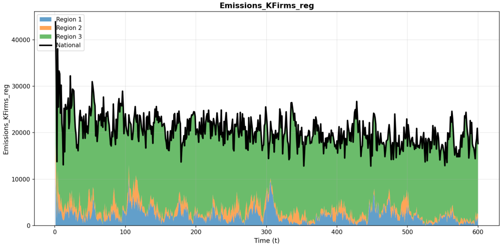


Figure: K firm Emissions

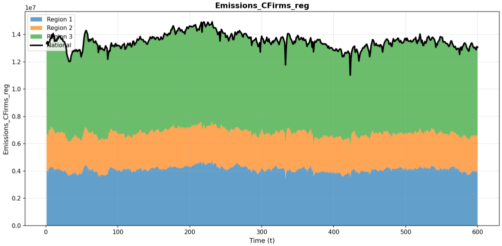


Figure: C firm Emissions

Sprint 2: Regionalising Emissions

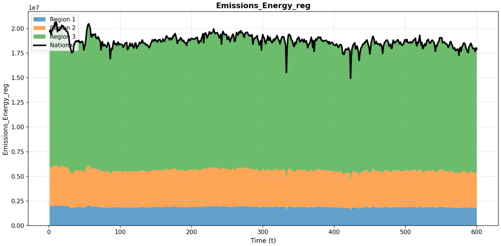


Figure: Energy Sector Emissions

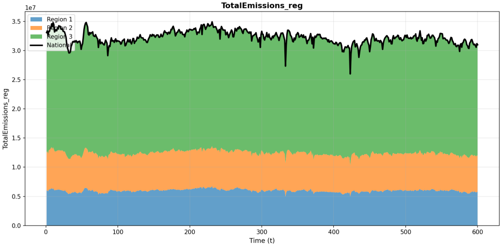


Figure: Total Emissions

Sprint 2: Regional Accounting of GDP

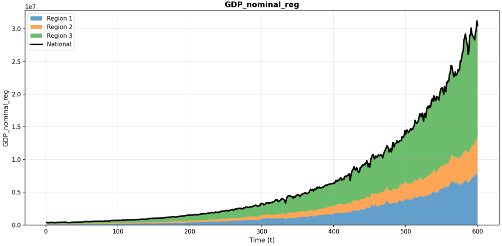


Figure: Nominal GDP

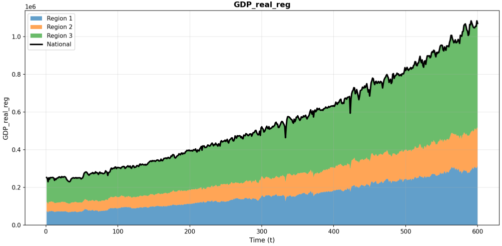


Figure: Real GDP

Regional Shocks

Sprint 3: Climate Shock Channels in Reissl et al. (2025)

- All firms in the region receive uniform shocks `flag_uniform_shock = 1`
  1. Shocks to productivity affecting the characteristics of capital vintages `flag_prodshocks1`
  2. Shocks to productivity (not affecting characteristics of capital vintages) `flag_prodshocks2`
  3. Energy Capacity both green and brown `flag_encapshocks`
  4. Shock to labour population `flag_popshocks`
  5. Shock to aggregate consumption demand `flag_demandshocks`
  6. Shock to C-Firms capital stock `flag_capshocks`
  7. Output shocks to C & K Firms `flag_outputshocks`
  8. Shock to C-Firms inventories `flag_inventshocks`
  9. Shock to R & D of K Firms `flag_RDshocks`

Sprint 3: Climate shocks in DSK

Climate shocks are modelled as stochastic shocks drawn from temperature-dependent Beta distributions in Reissl et al. (2025).
For each shock channel $s \in \{1, \ldots, S\}$ and period t:

$$\varepsilon_{s,t} \sim \mathrm{Beta}\left(a_{s,t},\ b_{s,t}\right), \qquad \varepsilon_{s,t} \in (0,1)$$

The shape parameters evolve with global temperature T_t:

$$a_{s,t} = a_{0,s}\left[1 + \ln\left(\frac{T_{t-1}}{T_0}\right)\right]^{\theta_s^{(1)}}$$

$$b_{s,t} = b_{0,s}\left(\frac{T_0}{T_{t-1}}\right)^{\theta_s^{(2)}}$$

- $a_{0,s}, b_{0,s}$ are baseline shape parameters,
  - $a_{s,t}$ governs mean/median of damages,
  - $b_{s,t}$ governs right tail thickness (extreme events),
- $\theta_s^{(1)}$ controls increase in typical damages,
- $\theta_s^{(2)}$ controls amplification of extremes,
- T_t is global mean temperature, T_0 is reference level.
- Climate shocks affect real economic variables (e.g. X) multiplicatively:

$$X_{i,t} = X_{i,t-1}\left(1 - \varepsilon_{s,t}\right)$$

Sprint 3: Parameters of Climate shock β distribution

1. **Global temperature** (T_t) is common to all regions
2. **Shock exponent $\theta_s^{(1)}$ and baseline severity parameters $a_{0,s}$** have regional values but are kept same common across regions as they reflect physical damage sensitivities (how temperature translates into damage)
3. **Baseline exposure $(b_{0,r})$**: Region-specific right-tail thickness capturing structural exposure to extreme events (e.g. coastal vs inland regions).
4. **Extreme-event amplification $(\theta_r^{(2)})$**: Region-specific scaling of tail risk as temperature rises, allowing extreme damages to accelerate faster in some regions.

Sprint 3: Regionalisation of climate shock channels

- In the regional extension, climate shocks are generated **at the regional level** and propagated to all firms located in that region.
- For each shock channel $s \in \{1, \dots, S\}$, region r, and period t:

$$\varepsilon_{s,r,t} \sim \mathrm{Beta}(a_{s,r,t},\, b_{s,r,t})$$

- The realised regional shock $\varepsilon_{s,r,t}$ is then broadcast to all firms (or specified channel) i in region r:

$$X_{i,t} = X_{i,t-1}(1 - \varepsilon_{s,r,t}), \qquad i \in r$$

- This mechanism is identical to Reissl et al. (2025).

Sprint 3: Regionalisation of climate shock channels

Regional Climate Shocks

- At each period, we draw one climate shock per region and per channel, driven by a common global temperature path and applied uniformly to all firms in the region.
- Regions differ in how strongly specific climate shock types hit (damages) and scale with warming, but all regions share the same global climate forcing.

Verification

Sprint 3: Verification

- Implementation of regional climate shock dynamics is verified by reproducing following climate-shock scenarios from Reissl et al. (2025).
  1. Baseline – No Shocks
     `flag_shockexperiment = 0;`
  2. Capital Stock Shocks (CS)
     `flag_shockexperiment = 1; flag_capshocks = 1`
  3. Labour Productivity Shocks (LP) + Energy Efficiency Shocks (EF)
     `flag_shockexperiment = 1; flag_prodshocks1 = 6`
- These scenarios are simulated separately on following models
  1. Reissl et al. (2025) DSK
  2. Regionalised DSK
- This ensures consistency with established agent-based climate shock dynamics.

Sprint 3: Verification | Climate shock parameters

Reissl et al. (2025)

```
...
"climshockparams": [
  {
    "nshocks": 9,
    ...
    "a2_0": 1,
    "b2_0": 100
    ...
    "a6_0": 1,
    "b6_0": 100,
    ...
    "a8_0": 1,
    "b8_0": 100,
    ...
    "shockexponent2_1": 3,
    "shockexponent2_2": 8,
    ...
    "shockexponent6_1": 3,
    "shockexponent6_2": 8,
    ...
    "shockexponent8_1": 3,
    "shockexponent8_2": 8,
    ...}
    ]
...
```

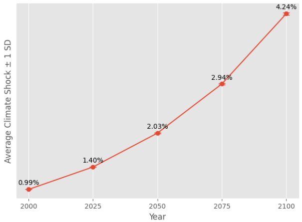
Regionalised SFC-DSK

```
...
"climshockparams": [
  {
    "nshocks": 9,
    ...
    "a2_0": [1, 1, 1],
    "b2_0": [100, 100, 100],
    ...
    "a6_0": [1, 1, 1],
    "b6_0": [100, 100, 100],
    ...
    "a8_0": [1, 1, 1],
    "b8_0": [100, 100, 100],
    ...
    "shockexponent2_1": [3, 3, 3],
    "shockexponent2_2": [8, 8, 8],
    ...
    "shockexponent6_1": [3, 3, 3],
    "shockexponent6_2": [8, 8, 8],
    ...
    "shockexponent8_1": [3, 3, 3],
    "shockexponent8_2": [8, 8, 8],
    ...}
    ]
...
```

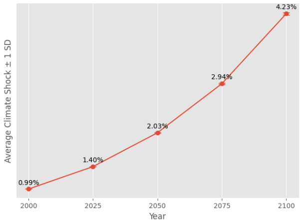
Sprint 3: Verification | Comparing Average Shock Values

Capital Stock Shock

DSK

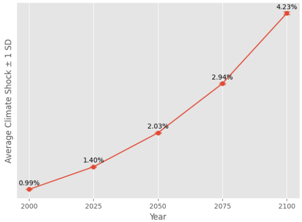


REGIONAL DSK

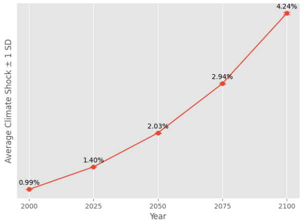


Labour Productivity & Energy Efficiency Shocks

DSK



REGIONAL DSK



Sprint 3: Verification | Comparing Average Shock Values

Scenario	Year	Mean (SFC-DSK)	Mean (Regional DSK)	$\lvert\Delta\rvert$
Labour Productivity (CS)				
	2000	0.0072	0.0072	$< 10^{-4}$
	2025	0.0103	0.0103	$< 10^{-4}$
	2050	0.0149	0.0149	$< 10^{-4}$
	2075	0.0216	0.0216	$< 10^{-4}$
	2100	0.0307	0.0307	$< 10^{-4}$
Energy Efficiency (LP+EF)				
	2000	0.0072	0.0072	$< 10^{-4}$
	2025	0.0103	0.0103	$< 10^{-4}$
	2050	0.0149	0.0149	$< 10^{-4}$
	2075	0.0216	0.0216	$< 10^{-4}$
	2100	0.0308	0.0307	10^{-4}

Notes: Reported values correspond to the Monte Carlo average of the mean shock size at selected benchmark years. Absolute differences between the baseline model and the regional SFC–DSK model with homogeneous shocks are on the order of 10^{-4} or smaller. Shock distributions are therefore identical across model structures.

Verification | Comparing Climate Damages

Here we compare climate damages between Reissl et al. (2025) and Regional DSK model with homogenous regions i.e. $R1 : R2 : R3 : R4 = 0.25 : 0.25 : 0.25 : 0.25$

	No Shocks	CS Shock	LP+EF Shock
Real GDP (ln)	0.9995 (0.9463)	0.9998 (0.3257)	1.0007 (-1.4043)
Bad debt over nominal GDP	1.0000 (0.0000)	1.0133 (-0.6202)	1.0164 (-0.7721)
Number of C-Firms failure	1.0000 (0.0000)	1.0043 (-0.7577)	1.0016 (-0.2554)
Endogenous Emissions	1.0000 (0.0000)	0.9980 (0.0929)	1.0058 (-0.2738)

Entries report normalized macroeconomic performance of the regional model (homogenous regions) relative to Reissl et al. (2025). Values above (below) one indicate amplification (attenuation). All values refer to Monte Carlo simulation runs of size 50. T-statistics for H_0: equal means are reported in parentheses.

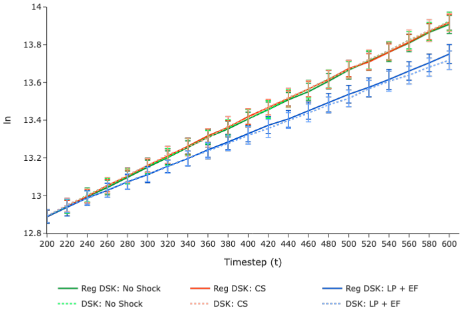
Verification | Comparing Sectoral Net-Worths

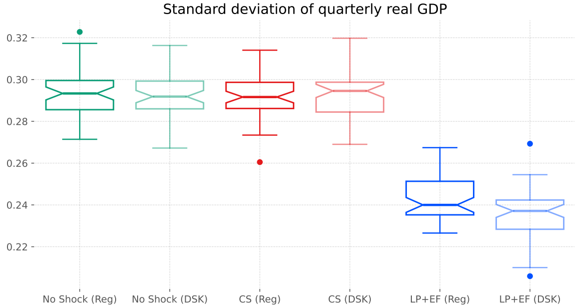
Change in sectoral net-worth under climate shocks between Reissl et al. (2025) and Regional DSK model with homogenous regions i.e. $R1 : R2 : R3 : R4 = 0.25 : 0.25 : 0.25 : 0.25$

▸ △ Sectoral Net Worth	CS Shock	LP+EF Shock
K-Firms	0.9905 (0.3036)	1.0875 (-2.4495)
C-Firms	1.0023 (-0.1451)	1.0308 (-1.8663)
Households	0.9978 (0.1529)	1.0149 (-1.2475)
Banks	0.9967 (0.2645)	0.9923 (0.5480)
Energy	1.0000 (-0.0352)	0.9902 (0.3462)
Government	0.9954 (-0.3377)	0.9846 (-1.2336)

Entries report ratio of normalized net-worth of the sectors relative to baseline (no shock) for regional DSK and Reissl et al. (2025). Values above (below) one indicate amplification (attenuation). All values refer to Monte Carlo simulation runs of size 50. T-statistics for H_0: equal means are reported in parentheses.

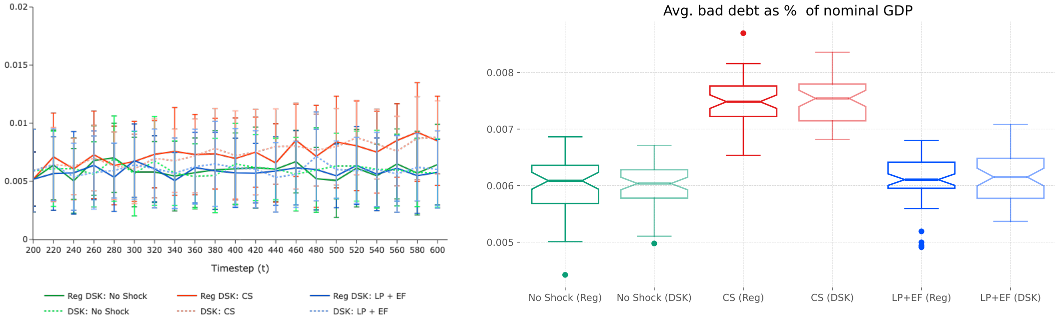
Verification | GDP





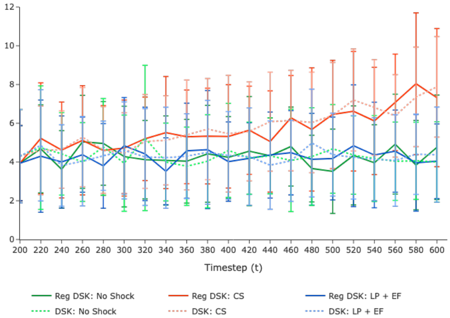
Entries report macroeconomic performance of the regional DSK model (homogenous regions) relative to Reissl et al. (2025). All values refer to Monte Carlo simulation runs of size 50.

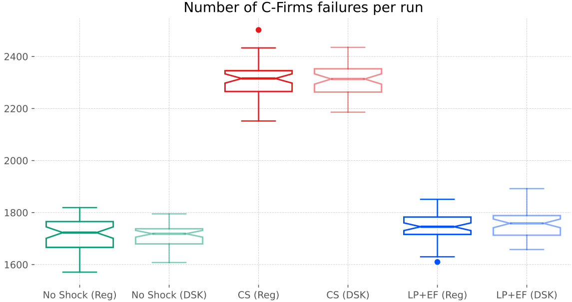
Verification | Bad Debt / Nominal GDP



Entries report macroeconomic performance of the regional DSK model (homogenous regions) relative to Reissl et al. (2025). All values refer to Monte Carlo simulation runs of size 50.

Verification | Exiting C-Firms

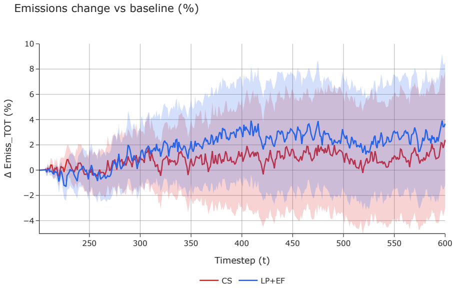




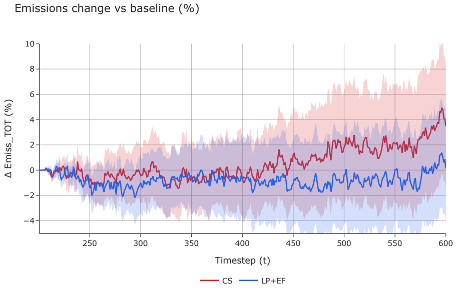
Entries report macroeconomic performance of the regional DSK model (homogenous regions) relative to Reissl et al. (2025). All values refer to Monte Carlo simulation runs of size 50.

Verification | Change in Sectoral Net-Worth Under Shocks

DSK Reissl et al. (2025) under CS and LP+EF



Regional DSK under CS and LP+EF

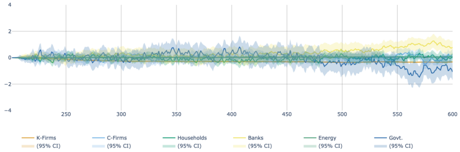


Verification | Change in Sectoral Net-Worth Under Shocks

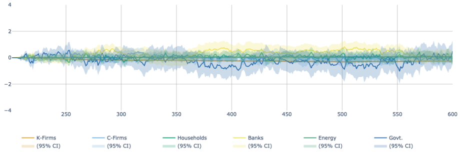
DSK Reissl et al. (2025) under CS

Regional DSK under CS

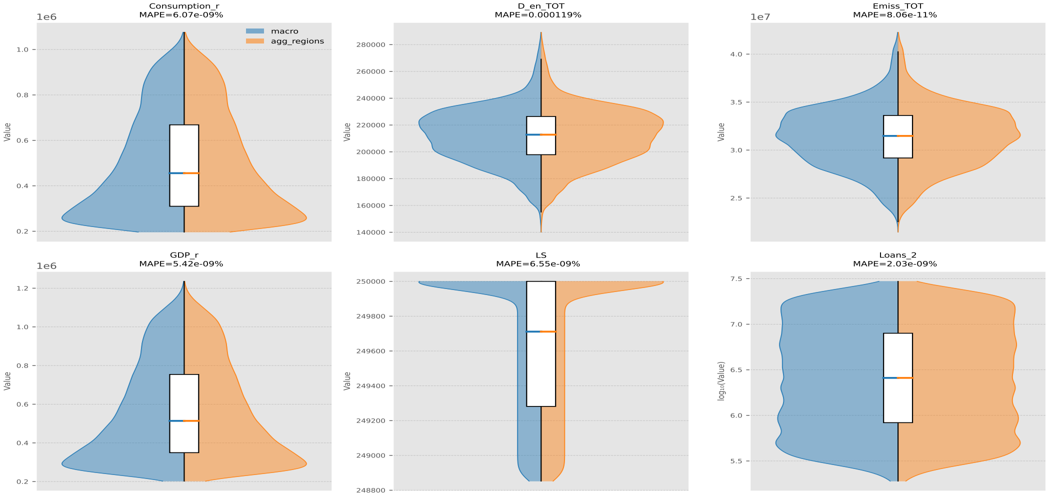
DSK Reissl et al. (2025) under LP+EF



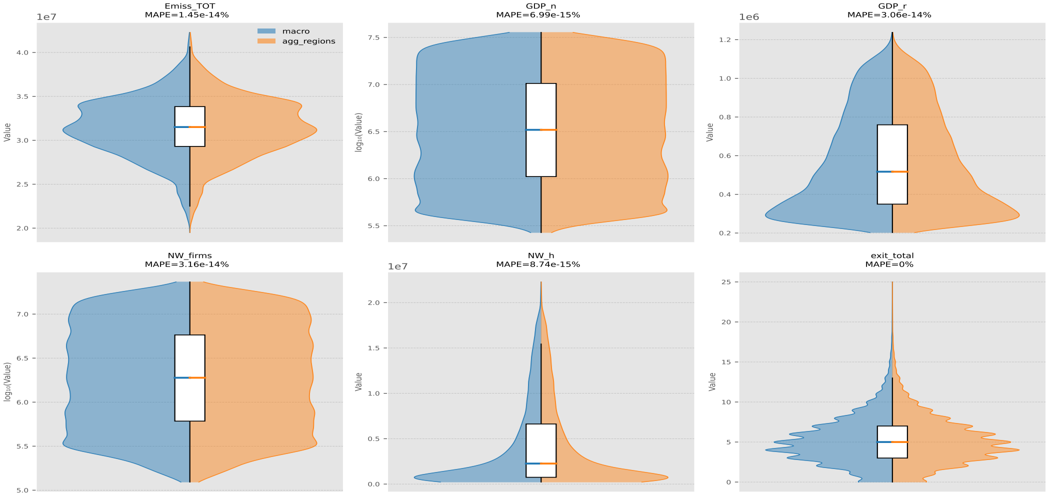
Regional DSK under LP+EF



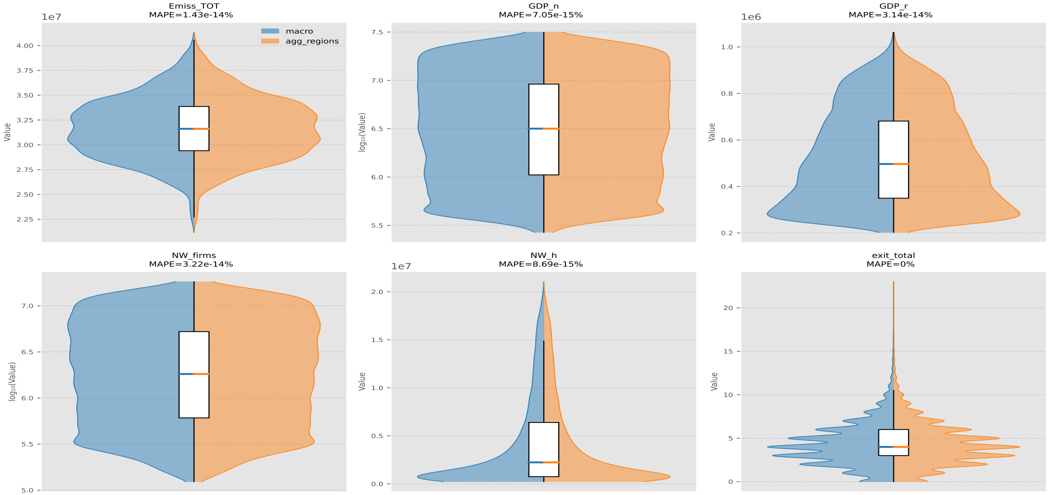
Verification | Regional Consistency Under Baseline (No Shock)



Verification | Regional Consistency Under Baseline (CS)



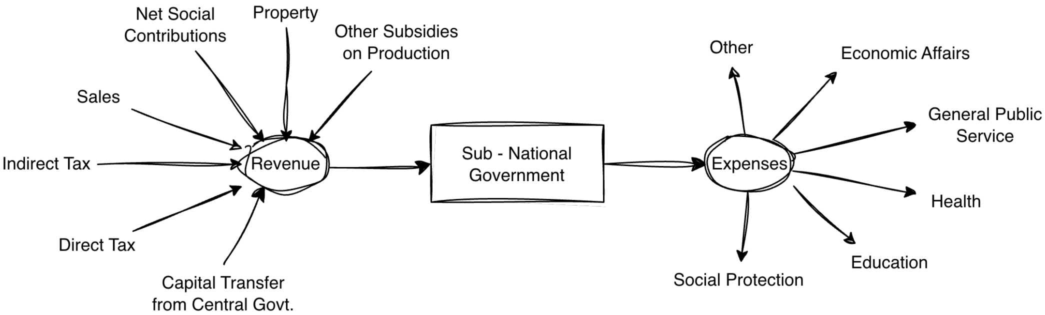
Verification | Regional Consistency Under Baseline (LP+EF)



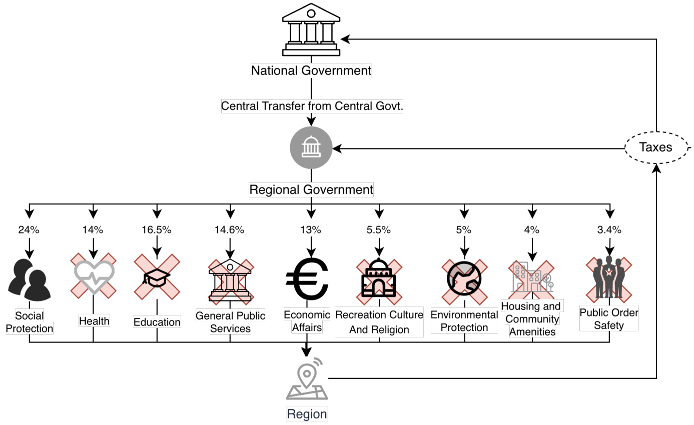
Regional Govt.

Conceptualising Sub-National Government in European Context

- Each country creates its own arrangement for dividing responsibilities between regional and local entities which can be collectively called as sub-national governments (Halaskova et al., 2021)
- Expenses and revenues of sub-national government can be calibrated using subnational government finances from ECB Olsson and Catz (2023)



Regional Govt. Revenue & Expenditure




```
RowVector Taxes_1;     // Taxes on profits paid by K-firms
RowVector Taxes_2;     // Taxes on profits  paid by C-firms
double Taxes_h;        // Taxes on income (wages) paid by households
RowVector Taxes_b;     // Taxes on profits paid by banks
RowVector Taxes_CO2_1; // Taxes on CO2 paid by K-firms
RowVector Taxes_CO2_2; // Taxes on CO2 paid by C-firms
double Taxes_CO2_e;    // Taxes on CO2 paid by energy sector
```

Regional Government Revenue Design

$$TS_{r,t} = \tau_r^{share}\left(T_{r,t}^{hh} + T_{r,t}^{C} + T_{r,t}^{K}\right) \quad (1)$$

$$GRANTPOOL_t = \bar{\gamma}\, REV_{CentralGovt,t} \quad (2)$$

$$GT_{r,t}^{base} = \omega_r\, GRANTPOOL_t \quad (3)$$

$$GT_{r,t}^{topup} = \max\left\{0, SP_{r,t}^{plan} - \left(TS_{r,t} + GT_{r,t}^{base}\right)\right\} \quad (4)$$

$$GT_{r,t} = GT_{r,t}^{base} + GT_{r,t}^{topup} \quad (5)$$

$$REV_{rg,r,t} = TS_{r,t} + GT_{r,t} \quad (6)$$

- $TS_{r,t}$: Region's share of centrally collected taxes
- $GT_{r,t}^{base}$: Standard intergovernmental transfer from grant pool
- $GT_{r,t}^{topup}$: Additional transfer ensuring minimum welfare provision
- $REV_{rg,r,t}$: Total regional government revenue
- National government collects all taxes and redistributes to regions
- Grant pool bounded: $\bar{\gamma} \in [0,1]$
- Allocation weights: $\sum_r \omega_r = 1$
- Welfare guarantee: $REV_{rg,r,t} \geq SP_{r,t}^{plan}$

Regional Government Expenditure Design

$$EXP_{rg,r,t} = SP_{r,t} + EA_{r,t}$$

Economic Affairs

$$EA_{r,t} = \max\left\{0, \quad REV_{r,t} - SP_{r,t)}\right\}$$

- $EA^{act}_{r,t} \rightarrow$ additional demand to regional C-firms
- For each C-firm in region r, $EA_{Cfirm,r,t} = \frac{EA_{r,t}}{N^{Cfirm}_{r,t}}$
- $Demand^{total}_{Cfirm,r,t} = Demand^{private}_{Cfirm,r,t} + EA_{Cfirm,r,t}$
- Government commits spending in t, but actual transactions occur in $t+1$
- Includes infrastructure, adaptation, regional demand stimulus

Social Protection

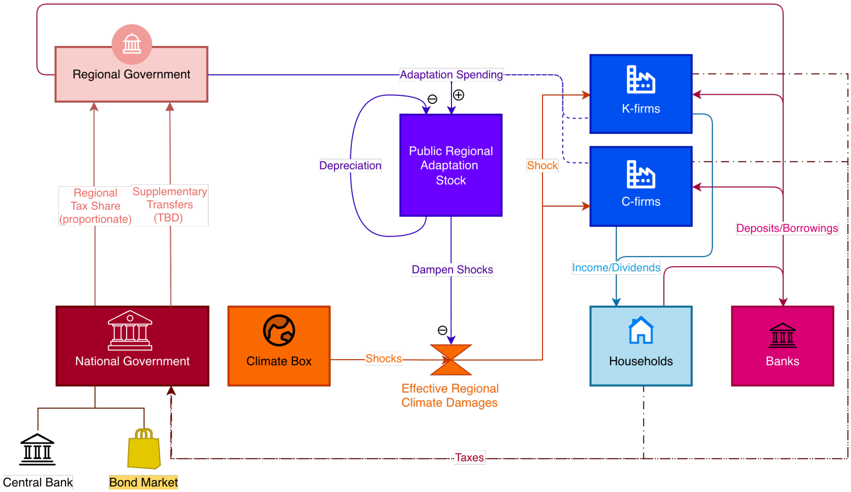
$$SP_{r,t} = \sigma_r \; Wages_{r,t}$$

- σ_r: social protection proportion
- Implemented as transfers to households

Regional Govt

TODO: Include reg-gov graphs

Regional Adaptation Conceptualisation



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